VLLM 10k Context 吞吐测试报告

- 测试时间: 2026-03-30 14:58:41
- Conda 环境: `xtyAgent`
- 基准接口: `http://127.0.0.1:62272/v1/completions`
- 模型名: `Qwen3-32B`
- 输入长度: 约 `10001` token
- 输出长度: `1024` token
- 并发集合: `5 10 20 40 80 160`
- 客户端 timeout: `180s`
- 请求数策略: `max(并发数, 10)`

结论

- 在 `1w` 输入负载下，最高完全稳定并发点是 `40`，成功吞吐约 `0.34` req/s，总 token 吞吐约 `3727.55` tok/s。
- 并发 `80` 和 `160` 都是 `100% timeout`，说明在当前 `180s` 超时预算下，系统容量拐点已经落在 `40` 与 `80` 之间。
- 非 timeout error 仍然是 `0`，失败全部来自排队/推理时间过长导致的 timeout。

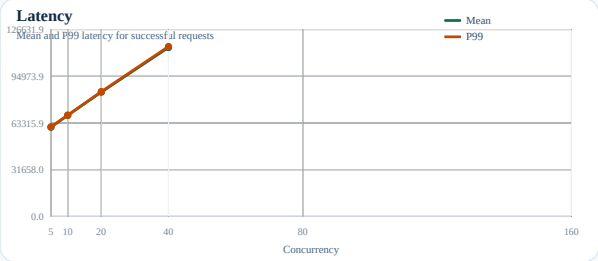
汇总表

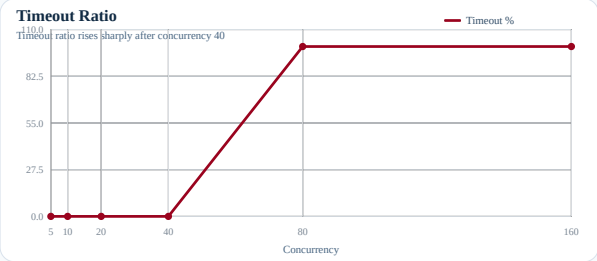
并发数	请求数	成功	Timeout	Mean(ms)	P99(ms)	Timeout%	Success req/s	Total tok/s
5	10	10	0	60502.36	60650.28	0.00%	0.08	855.01
10	10	10	0	68501.00	68666.66	0.00%	0.14	1534.78
20	20	20	0	84188.99	84587.07	0.00%	0.23	2512.57
40	40	40	0	114460.30	115119.90	0.00%	0.34	3727.55
80	80	0	80	N/A	N/A	100.00%	0.00	0.00
160	160	0	160	N/A	N/A	100.00%	0.00	0.00

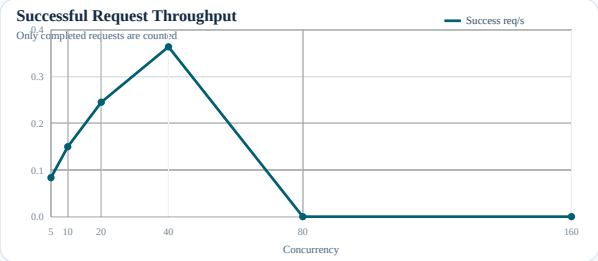
图表

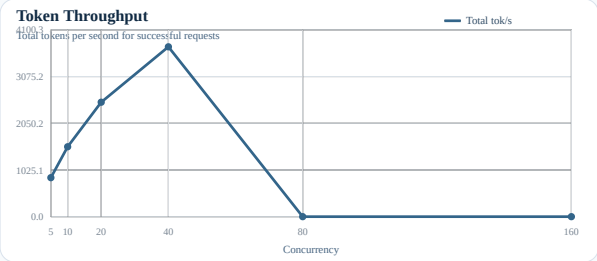
VLLM Benchmark Dashboard: 10k Input Context

Date: 2026-03-30 14:58:41 | Endpoint: http://127.0.0.1:62272/v1/completions | Payload: 10000 input / 1024 output tokens









10k Context Summary Table

并发	成功	timeout%	Mean(ms)	P99(ms)	req/s	tok/s
5	10/10	0.00%	60502.36	60650.28	0.08	855.01
10	10/10	0.00%	68501.00	68666.66	0.14	1534.78
20	20/20	0.00%	84188.99	84587.07	0.23	2512.57
40	40/40	0.00%	114460.30	115119.90	0.34	3727.55
80	0/80	100.00%	N/A	N/A	0.00	0.00
160	0/160	100.00%	N/A	N/A	0.00	0.00

Key Findings

1. Concurrency 40 is the highest fully stable point under 10k input load.
2. Concurrency 40 reaches 0.34 req/s and 3727.55 total tok/s.
3. Concurrency 80 and 160 both hit 100% timeout at the 180s client limit.
4. Non-timeout error rate remains 0%; failures are pure timeout saturation.
5. Compared with the 2k-input run, the practical concurrency ceiling shifts down markedly.

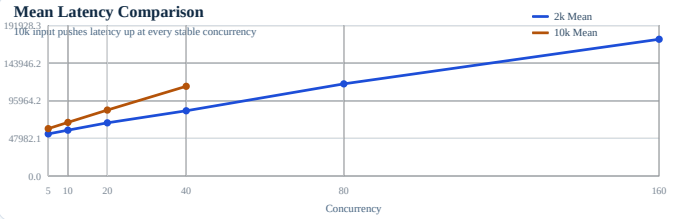
对比图

VLLM Benchmark Comparison: 2k vs 10k Input Context

Same endpoint and output length; comparison date 2026-03-30 14:58:41

Mean Latency Comparison

10k input pushes latency up at every stable concurrency

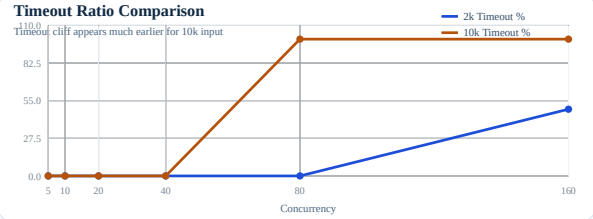


2k vs 10k Summary

并发	2k Mean	10k Mean	2k Timeout%	10k Timeout%	2k re
5	53723.67	60502.36	0.00%	0.00%	0.0
10	58459.94	68501.00	0.00%	0.00%	0.1
20	67825.09	84188.99	0.00%	0.00%	0.2
40	83235.00	114460.30	0.00%	0.00%	0.4
80	117575.98	N/A	0.00%	100.00%	0.6
160	174480.29	N/A	48.75%	100.00%	0.4

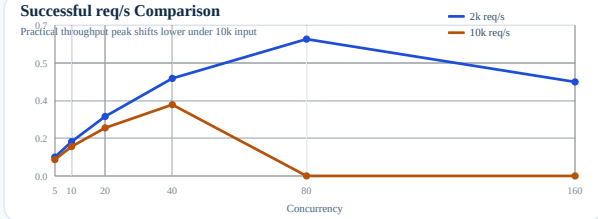
Timeout Ratio Comparison

Timeout cliff appears much earlier for 10k input



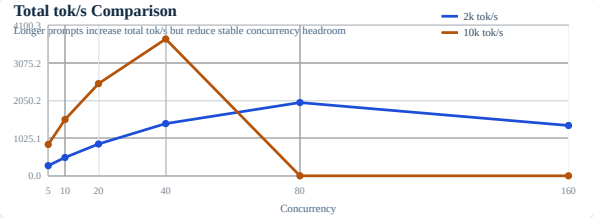
Successful req/s Comparison

Practical throughput peak shifts lower under 10k input



Total tok/s Comparison

Longer prompts increase total tok/s but reduce stable concurrency headroom



Interpretation

1. 2k input remains stable through concurrency 80, while 10k input saturates before 80.
2. At concurrency 40, mean latency rises from 83.24s to 114.46s when input grows from 2k to 10k.
3. At concurrency 80, 2k still has 0% timeout and 0.65 req/s, but 10k becomes 100% timeout.
4. 10k input improves total successful tok/s at low stable concurrency, but loses queueing headroom quickly.
5. For production, 10k input should keep concurrency at or below 40 under this timeout budget.